

1. What is the smallest positive integer that is divisible by each of the first five counting numbers (1, 2, 3, 4, and 5)?

(4 marks)

2. What is the largest three-digit number that is the sum of the squares of two consecutive even natural numbers?

(4 marks)

3. How many perfect squares exist between one hundred thousand and one million?

(4 marks)

4. What reflex angle is formed by the minute hand and the hour hand of a clock at 3:45?

(4 marks)

5. Try to solve this riddle:
My son is five times as old as my daughter, and my wife is five times as old as my son. I am twice as old as my wife, and my grandmother, who is as old as all of us put together, is 81 years old today. How old is my son?

(4 marks)

6. In a survey of 756 high school students, 70 percent more students preferred cell phones to landline phones. How many students preferred landline phones?

(4 marks)

7. The faces of a die are marked with the numbers 1 through 6. Each corner of the die is assigned a *vertex number* that is equal to the sum of the numbers on the faces that meet at that vertex. What is the sum of all the *vertex numbers*?

(5 marks)

8. What is the value of $1 + (1 + 2) + (2 + 3) + (3 + 4) + \dots + (2013 + 2014)$?

(5 marks)

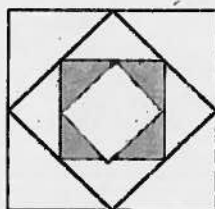
9. The sum of two consecutive odd numbers is the square of the product of the first and fourth prime numbers. What are the two odd numbers?

(5 marks)

10. David signed a contract that says he must build 20 doghouses for the SPCA within one month. For every doghouse that David completes, he will receive R400; for every one that he fails to complete, he will be fined R100. At the end of the month, David received R4 000. How many doghouses did he build?

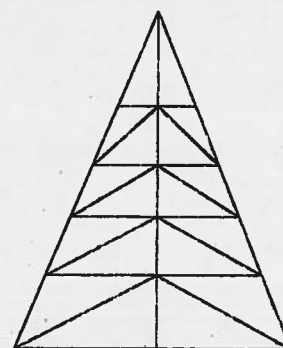
(5 marks)

11. The dart board shown is a 12 cm square. Each inner square is formed by joining the midpoints of the larger square. What is the probability that a randomly thrown dart that hits the dart board hits the shaded area? Express your answer as a common fraction.



(5 marks)

12. How many triangles are in the figure below?



(5 marks)

13. What is the remainder when 7^{2014} is divided by 5?

(5 marks)

14. List all the positive integers less than 20 000 which are both perfect squares and perfect cubes.

(5 marks)

15. Sarah wrote letters to her friends Abby, Betsy, Candice, and Delia. In how many different ways could she address the envelopes so that all four friends receive the wrong letter?

(5 marks)

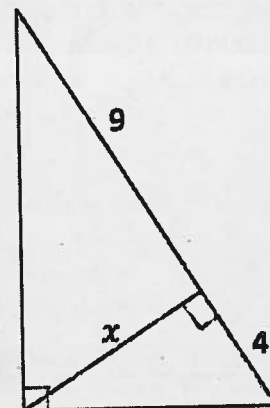
16. I have fewer than 100 baseball cards. When I divide the total number of cards by 4, then 3 cards are left over. When I divide the number of cards by 3, no cards are left over. When I divide the number of cards by 2, only 1 card is left over. But if I divide the total number of cards by 7, no cards are left over. How many baseball cards do I have?

(5 marks)

17. I started my two clocks at the same time. One is slow and loses 2 minutes every hour; the other one is fast and gains 1 minute every hour. How long will it take for the faster clock to be 1 hour ahead of the slower clock?

(5 marks)

18. Determine x :



(5 marks)

19. If n is a perfect square then what is the next perfect square greater than n ?

(6 marks)

20. If the digits k , m and n of the three digit number kmn satisfy $64k + 8m + n = 637$, then what is the number kmn ?

(6 marks)

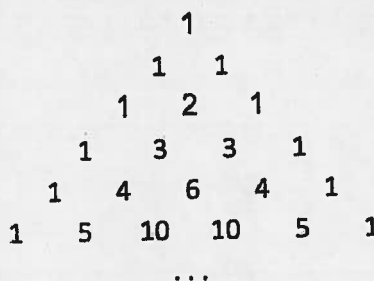
21. At the conclusion of a committee meeting, a total of 66 handshakes were exchanged. Assuming that each person was equally polite toward all the others, how many people were present?

(6 marks)

22. When 5 new girls joined a class, the percent of girls increased from 40% to 50%. What is the number of boys in the class?

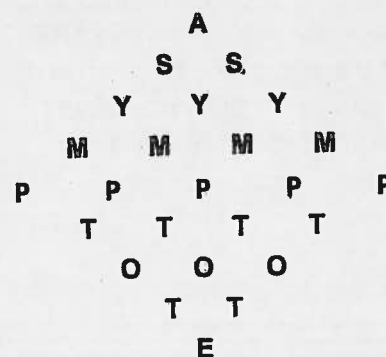
(6 marks)

23. Determine the 122nd number of Pascal's triangle given below:



(6 marks)

24. How many paths from top to bottom spell PATTERN?



(6 marks)

Total: 120 marks

Tour de Maths 2014 Junior Leg 2 - SJC

Memo

Marks/Quest	1	2	3	TOTALS
4	60	884	683	
	4	5	6	
4	112.5°	5	5	
	7	8	9	
5	84	4 056 196 or 2014 ²	97 and 99	
	10	11	12	
5	12	$\frac{1}{4}$	43	
	13	14	15	
5	4	1; 64; 729; 4096; 15625.	9	
	16	17	18	
5	63	20	6	
	19	20	21	
6	$(\sqrt{n} + 1)^2$	975	12	
	22	23	24	
6	15	15	70	
			TEAM TOTAL	

